

Student Support Dashboard Analytics for a Public High School District

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Abstract

Chronic absenteeism and school discipline measures are often reviewed separately, even though both can point to student support needs that matter for school improvement. This project uses public school-level data for Maine Township High School District 207 to build a small district analytics workflow for attendance, behavior, enrollment, subgroup review, data quality checks, and dashboard-ready reporting. Data were obtained from the Urban Institute Education Data Portal, which provides programmatic access to Common Core of Data and Civil Rights Data Collection records [1, 2, 3]. The primary hypotheses were that public records can be transformed into a useful school support monitoring layer, and that a transparent composite score can identify schools that warrant deeper local review without implying a causal effect of any program. In 2021 public records, the district chronic absenteeism rate was 34.1%, and suspension instances occurred at 4.6 per 100 students. Maine West High School ranked highest on the composite support-priority score, followed by Maine East High School and Maine South High School. The workflow is best interpreted as an early warning and dashboard design exercise, not as an evaluation of specific interventions.

1 Introduction

School districts rely on attendance, behavior, assessment, and enrollment data to decide where additional support may be needed. Chronic absenteeism is a useful example because it is common, measurable, and closely tied to student outcomes. The U.S. Department of Education describes chronic absenteeism as a major educational challenge, and Attendance Works emphasizes that missing 10% or more of school for any reason can place students at academic risk [4, 5]. Illinois also defines chronic absence in statute and expects schools to monitor attendance problems in a systematic way [7].

Attendance data alone are not enough. Schools also need behavior indicators, subgroup breakdowns, and data-quality checks so leaders can distinguish a true pattern from a reporting issue. Multi-Tiered System of Supports frameworks encourage schools to use data to connect universal, targeted, and intensive supports to student needs [8]. Evidence guidance from the Institute of Education Sciences also recommends using student data to guide instructional and support decisions, while recognizing that data use requires clear measures, timely reporting, and careful interpretation [9].

This project studies what can be learned from public, aggregate data. It does not contain student identifiers, course grades, intervention assignments, visit-level records, or case notes. That

constraint is important. Public data are useful for district-level monitoring, public accountability, and dashboard prototypes, but they should not be stretched into claims about individual students or intervention effectiveness. Student-level analysis would require local data governance, privacy protections, and clear compliance with the Family Educational Rights and Privacy Act [11].

2 Research Questions and Hypotheses

The analysis addresses three practical research questions:

1. Can public school data be integrated into a clean monitoring layer for school attendance, behavior, enrollment, subgroup review, and data-quality checks?
2. Do school-level attendance and behavior measures identify meaningfully different support-priority profiles across comprehensive high schools?
3. Which subgroup chronic absenteeism patterns should be flagged for deeper local review while avoiding overinterpretation of small public-data cells?

The hypotheses were:

H1. Public CCD and CRDC records can be transformed into a usable district monitoring layer with consistent school identifiers, transparent denominators, and dashboard-ready tables.

H2. A composite support-priority score based on chronic absenteeism, suspension instances, and student-need context will separate schools into distinct monitoring priorities, while preserving the underlying measures for interpretation.

H3. Subgroup chronic absenteeism rates will show notable variation within schools, but small-denominator and suppressed cells will require stability flags before the results are used for planning.

3 Data and Measures

Data were collected from the Urban Institute Education Data Portal API [1]. The school directory and enrollment files came from the Common Core of Data. Chronic absenteeism, subgroup enrollment, and discipline-instance measures came from the Civil Rights Data Collection, which is a federal source for public school civil rights indicators [3].

The district was identified using `leaid = 1724090`. The school-level analytic file retained four public school records: Maine East High School, Maine South High School, Maine West High School, and Ralph J Frost Academy. The three comprehensive high schools had positive latest-year CCD enrollment. Ralph J Frost Academy was retained in quality-control outputs, but rows with zero or missing enrollment were excluded from rate-based comparisons.

The main measures were enrollment, chronic absenteeism, suspension instances, student-need context, subgroup chronic absenteeism rates, and data-quality checks. Chronic absenteeism was calculated as the CRDC count of chronically absent students divided by CRDC enrollment. Suspension instances were scaled per 100 enrolled students. Student-need context used latest CCD free or

reduced-price lunch share. Subgroup review focused on 2021 CRDC race subgroup rates and applied a stability threshold of at least 40 enrolled students for the main subgroup review.

4 Methods

4.1 Data Extraction and Cleaning

The download script queried the Urban Institute Education Data Portal API for Illinois school records, then filtered locally to the district LEA identifier. This approach was used because not every API endpoint applied a local education agency name filter consistently. The cleaning script standardized school identifiers, converted negative or suppression-style public-data codes to missing values, and created analysis-ready CSV files for enrollment trends, chronic absenteeism, discipline instances, subgroup rates, dashboard KPIs, and data-quality checks.

Data quality was handled before rate calculation. A rate was created only when the denominator was positive and nonmissing. Rows with missing numerator, missing denominator, zero enrollment, or suppressed subgroup values were retained in the audit files but not treated as valid rate estimates. This is consistent with the general data-quality principle that definitions, completeness, and fitness for use should be checked before analytical reporting [10].

4.2 Statistical Analysis

The analysis is descriptive and diagnostic. It uses rates, per-100-student scaling, school-level ranks, subgroup comparisons, and standardized components. No inferential tests were used because the available public files are aggregate administrative records rather than a random sample, and because only a small number of schools were observed. Instead, the analysis emphasizes transparent denominators and interpretable effect sizes.

The composite support-priority score was calculated across schools with valid 2021 values. Three components were standardized with z-scores: chronic absenteeism rate, suspension instances per 100 students, and free or reduced-price lunch share. The final score used a 45% weight for chronic absenteeism, 35% for suspension instances per 100 students, and 20% for free or reduced-price lunch share. The weights prioritize direct support indicators while still retaining a small contextual component. This score is not a definitive performance rating. It is a dashboard triage measure that points to where a school team might begin a more detailed review.

4.3 Dashboard and SQL Layer

The reporting layer was built in SQLite using the processed CSV files. The SQL views cover district KPIs, school chronic absenteeism trends, subgroup equity review, behavior-support monitoring, MTSS priority ranking, and data-quality checks. Those views are intentionally simple. They use plain `SELECT`, filtering, ordering, and rate fields so the logic can be inspected by analysts and district stakeholders.

5 Results

5.1 District-Level Support Indicators

The latest CCD directory data showed three active comprehensive high schools with positive enrollment, totaling 6,272 students. In CRDC 2021 records, 2,137 of 6,335 students were chronically absent, producing a district chronic absenteeism rate of 34.1%. Suspension instances occurred at 4.6 per 100 students. These public indicators are high enough to justify routine dashboard review. They do not explain why students were absent, whether absences were excused, or which interventions students received. They do, however, provide a clear starting point for school-level monitoring.

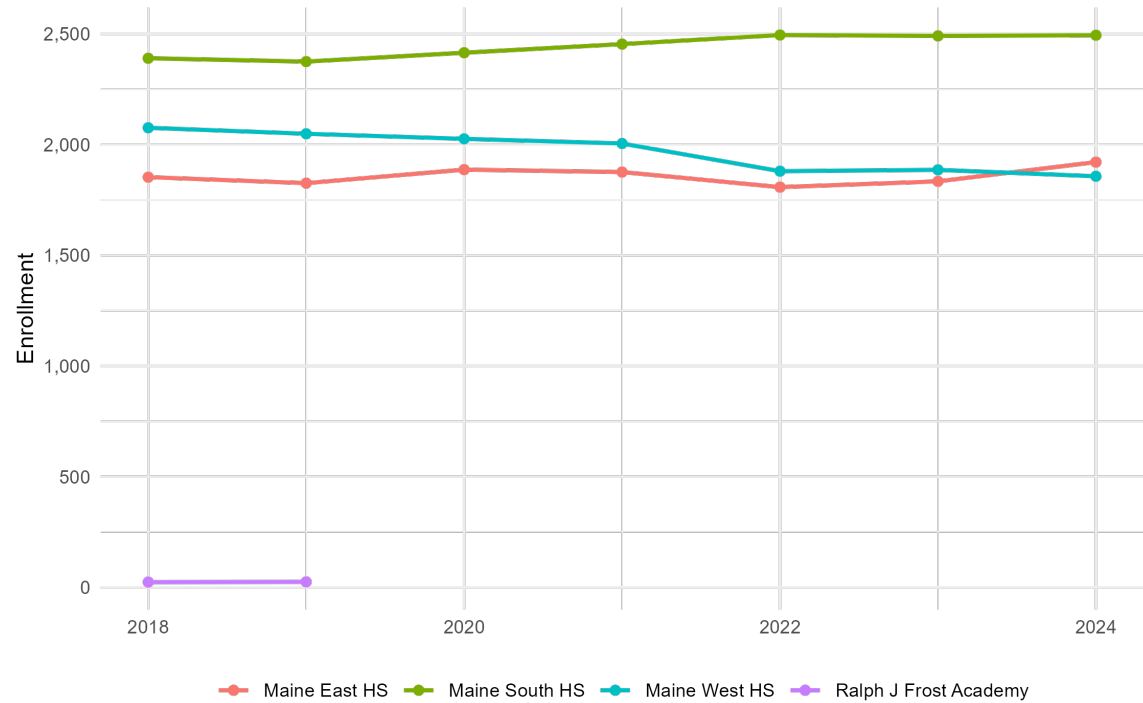


Figure 1: CCD enrollment trends by school, 2018 through 2024.

5.2 Chronic Absenteeism Increased Across the Comprehensive High Schools

Each comprehensive high school had a higher chronic absenteeism rate in 2021 than in 2020. Maine West increased from 31.6% to 43.7%, Maine East increased from 27.4% to 37.8%, and Maine South increased from 14.6% to 22.5%. The ordering was consistent in 2021: Maine West had the highest chronic absenteeism rate, followed by Maine East, then Maine South.

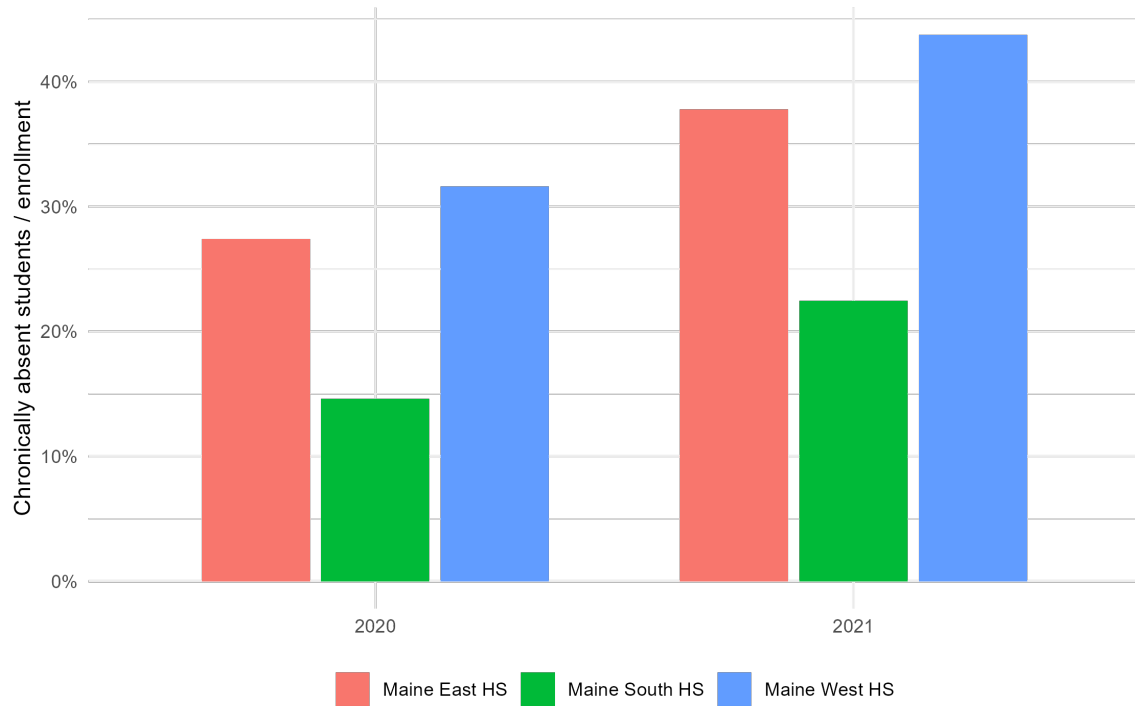


Figure 2: CRDC chronic absenteeism rate by school, 2020 through 2021.

5.3 Behavior-Support Measures Pointed to the Same Highest-Priority School

Suspension-instance rates also increased sharply in 2021. Maine West recorded 7.2 suspension instances per 100 students, Maine East recorded 5.2, and Maine South recorded 1.9. Because suspension instances can count multiple events for the same student, this measure should not be read as the percent of students suspended. It is a behavior-support activity rate that can help identify where further review may be useful.

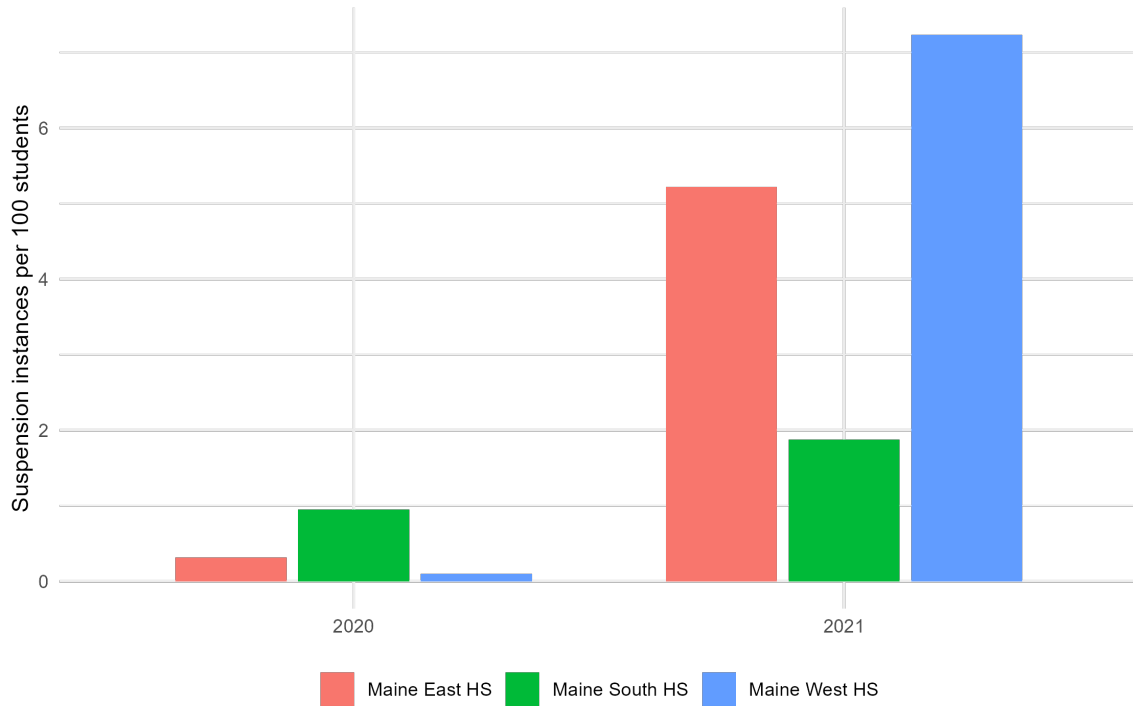


Figure 3: CRDC suspension instances per 100 students by school, 2020 through 2021.

5.4 Composite Priority Ranking Separates Schools Without Hiding the Inputs

The composite score ranked Maine West highest, followed by Maine East and Maine South. The ranking was not driven by a single input. Maine West had the highest chronic absenteeism rate and highest suspension-instance rate in 2021, while Maine East had the highest free or reduced-price lunch share. Maine South had lower values on all three components.

Rank	School	Enrollment	Chronic absence	Susp. per 100	FRPL share	Score
1	Maine West High School	2,005	43.7%	7.2	32.5%	0.77
2	Maine East High School	1,876	37.8%	5.2	37.6%	0.34
3	Maine South High School	2,454	22.5%	1.9	5.7%	-1.11

Table 1: Composite support-priority table for schools with valid 2021 public data.

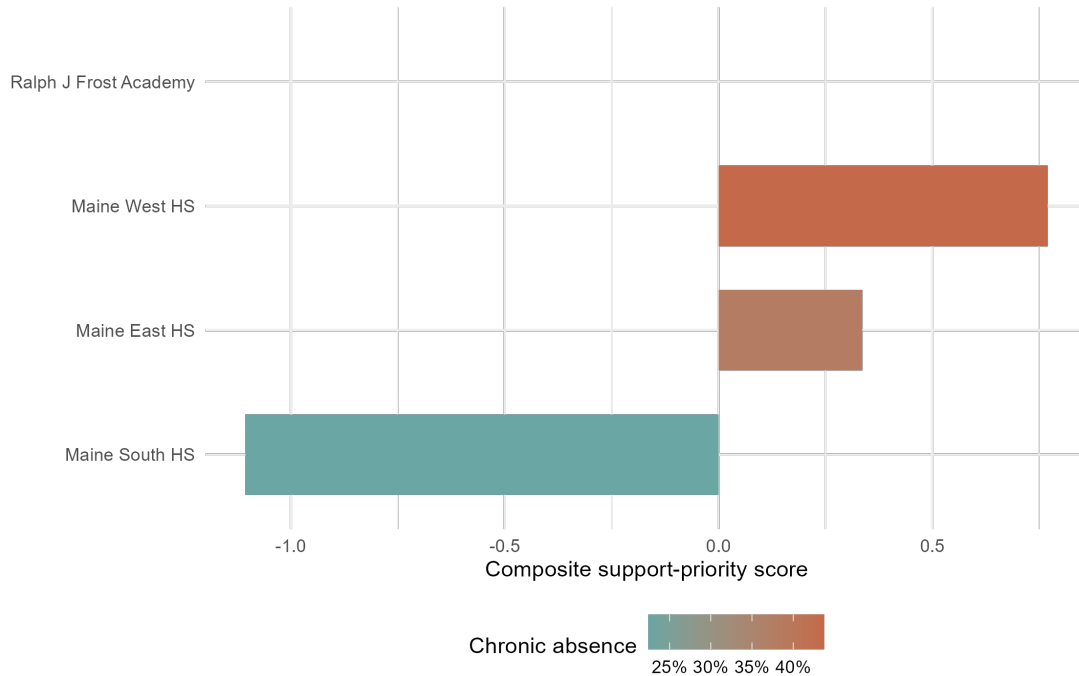


Figure 4: Composite support-priority score by school, 2021 public records.

5.5 Subgroup Review Requires Both Rates and Denominators

Subgroup chronic absenteeism rates varied within schools. In the 2021 stable subgroup review, Black and Hispanic students had chronic absenteeism rates above 50% at Maine West, and Black, Hispanic, and two-or-more-races groups were elevated at Maine East. At Maine South, Hispanic and two-or-more-races groups had higher rates than the schoolwide White subgroup, though the absolute pattern was lower than at Maine East and Maine West.

These comparisons should be interpreted with denominator context. Small groups can show unstable percentages, so the main chart retained only race subgroup rows with at least 40 enrolled students and nonsuppressed numerator/denominator values. Attendance Works recommends monitoring attendance with frequent, actionable data, but the monitoring process still needs care around definitions and cell size [6].

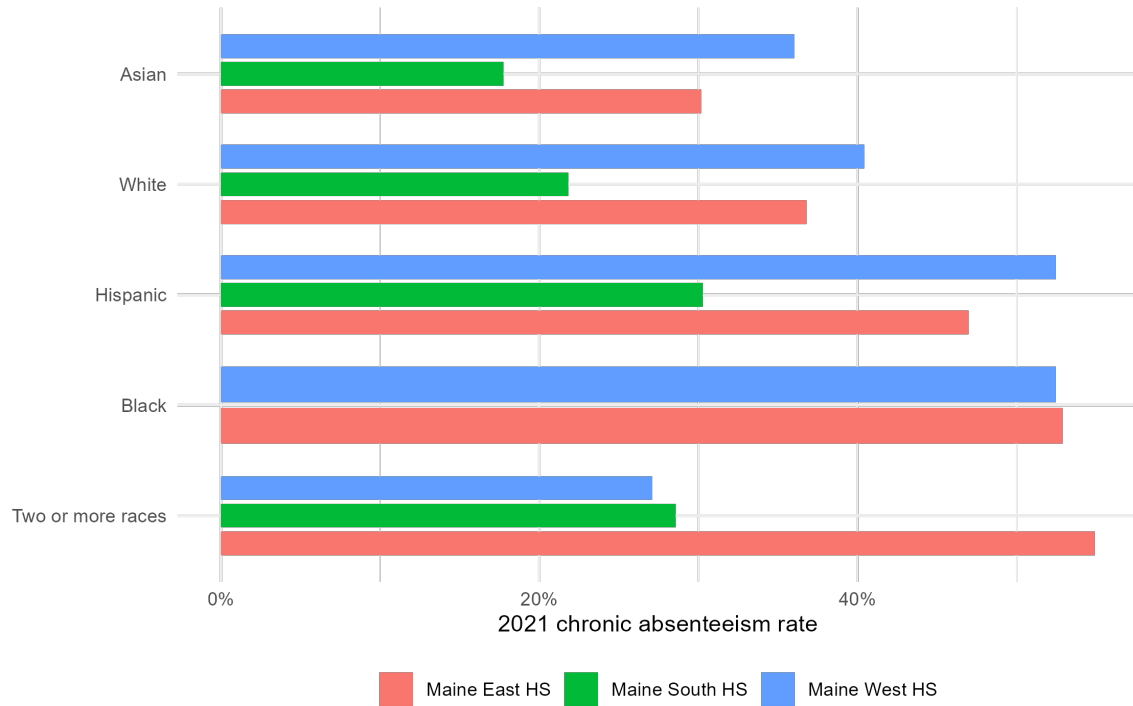


Figure 5: Stable race subgroup chronic absenteeism rates, CRDC 2021.

Table 2: Stable race subgroup chronic absenteeism review, CRDC 2021.

School				Subgroup	Enrollment	Chronically absent	Rate
Maine School	East	High	Asian		570	172	30.2%
Maine School	East	High	Black		87	46	52.9%
Maine School	East	High	Hispanic		462	217	47.0%
Maine School	East	High	Two or more races		51	28	54.9%
Maine School	East	High	White		666	245	36.8%
Maine School	South	High	Asian		107	19	17.8%
Maine School	South	High	Hispanic		261	79	30.3%
Maine School	South	High	Two or more races		56	16	28.6%
Maine School	South	High	White		1,983	433	21.8%

School		Subgroup		Enrollment	Chronically absent	Rate
Maine School	West	High	Asian	197	71	36.0%
Maine School	West	High	Black	61	32	52.5%
Maine School	West	High	Hispanic	865	454	52.5%
Maine School	West	High	Two or more races	48	13	27.1%
Maine School	West	High	White	757	306	40.4%

5.6 Data Quality Checks Are Part of the Analytic Result

The workflow produced five dashboard-ready processed tables. Data-quality checks found no missing school names in the chronic absenteeism records. Two total-rate rows had missing chronic absenteeism rates because the denominator was zero or missing. Eight small subgroup rows were suppressed from the main equity chart. These checks mattered because they prevented zero-enrollment or small-cell rows from being interpreted as stable rates.

Check	Value
CCD directory rows	28
CRDC chronic absenteeism rows	172
CRDC enrollment 2021 rows	128
Discipline total rows	8
Active latest schools	3
Missing school names in chronic records	0
Missing chronic-rate total rows	2
Small subgroup rows suppressed from main equity chart	8

Table 3: Data-quality checks generated by the analysis script.

6 Discussion

The project supports the first hypothesis. Public CCD and CRDC records were sufficient to build a district monitoring layer with school identifiers, attendance indicators, behavior-support measures, subgroup rates, quality checks, and SQL reporting views. This is a practical result because school analytics work often depends as much on data validation and common definitions as it does on statistical modeling.

The second hypothesis was also supported, within the limits of aggregate data. The composite score separated schools into distinct monitoring priorities. Maine West was the clearest high-priority school in the 2021 public data because both chronic absenteeism and suspension-instance rates were highest there. Maine East also warranted deeper review, especially because it combined

elevated chronic absenteeism with the highest student-need context measure. Maine South had lower values on the dashboard indicators, though it still had subgroup patterns worth monitoring.

The third hypothesis was partly supported. Subgroup rates varied meaningfully, but the public-data structure required stability controls. A high subgroup rate can be a real signal, a small-denominator artifact, or both. The safest operational use is to treat subgroup rates as prompts for local review rather than final conclusions.

7 Limitations

The analysis has several important limits. First, it uses public aggregate data, not student-level longitudinal records. It cannot identify whether the same students experienced chronic absenteeism and discipline events. Second, suspension instances are events, not unique students, so they measure school behavior-support activity rather than the share of students disciplined. Third, the CRDC files used here cover 2020 and 2021, years affected by pandemic-era attendance and schooling disruptions. Fourth, the analysis does not include local intervention assignment, fidelity, dosage, or outcome data. As a result, the score cannot evaluate intervention effectiveness.

Several useful extensions would require local data access. A district could link daily attendance, course performance, assessment results, intervention rosters, service minutes, and student support notes under approved privacy rules. With those records, the dashboard could move from school-level monitoring to student-level early warning, intervention matching, and outcome evaluation.

8 Conclusion

Public education data can support a clear, auditable first layer of district analytics. In this project, CCD and CRDC records were transformed into dashboard-ready school trends, subgroup views, behavior-support measures, data-quality checks, and SQL reporting views. The 2021 public data highlighted Maine West and Maine East as schools that would benefit from deeper attendance and behavior-support review, while preserving the underlying rates and denominators needed to interpret that result. The strongest next step would be to connect this public monitoring layer to local, privacy-governed student information system data so school teams can evaluate interventions and support decisions at a more actionable grain.

References

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